

Text as Pieces

Project: Transparent Book Recommender

1. Your mission

Your learning target

I can encode and decode text under an explicit toy vocabulary.

I can explain how segmentation changes sequence length.

I can handle unsupported characters without silently deleting information.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Feedback Loops. Give an example and a mistake that the idea helps you avoid.

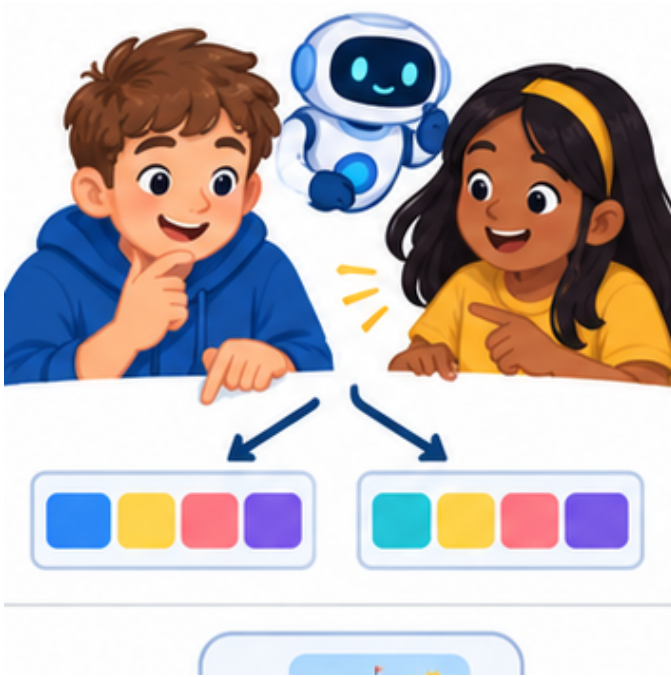
Without opening the old lesson, explain one key idea from Nearest Neighbor Reasoning. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Sampling and Representation. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 25: Text as Pieces

Language systems process tokens and patterns rather than whole human meanings.



THINKER MISSION

Tokenize tricky phrases and compare alternate splits.

Thinker name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

What information conditions a next-token distribution?

This week's end goal

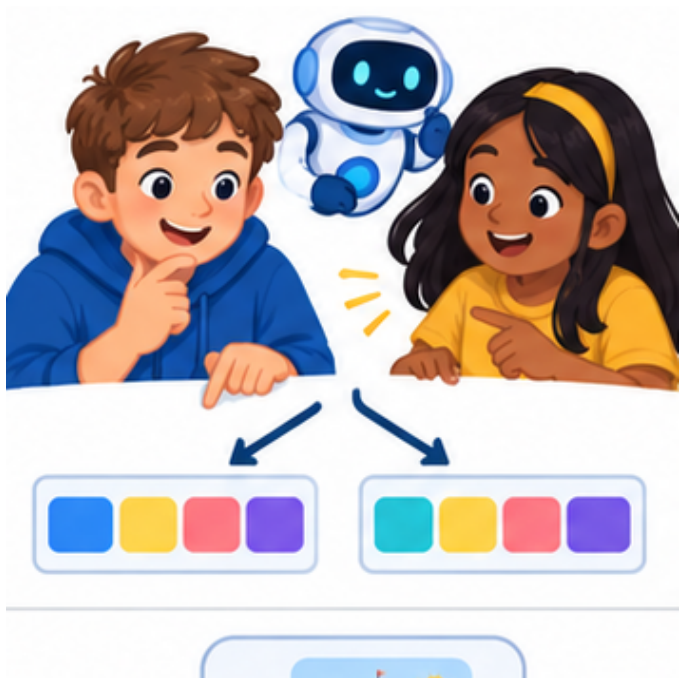
By the end, I can encode and decode text using a specified vocabulary and matching rule.

2. Notice the evidence

PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use text as pieces to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

Write or draw your reasoning and evidence.

My prediction and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

Tokens need not be words. A classroom tokenizer can use an exact, case-sensitive vocabulary and greedily choose the longest matching token from left to right. This is a toy rule, not a claim about every production tokenizer. Define IDs: 0=end, 1=cat, 2=c, 3=a, 4=t, 5=s, 6=space, 7=sat. The space token represents one literal space. The end token is a special terminator, not ordinary text.

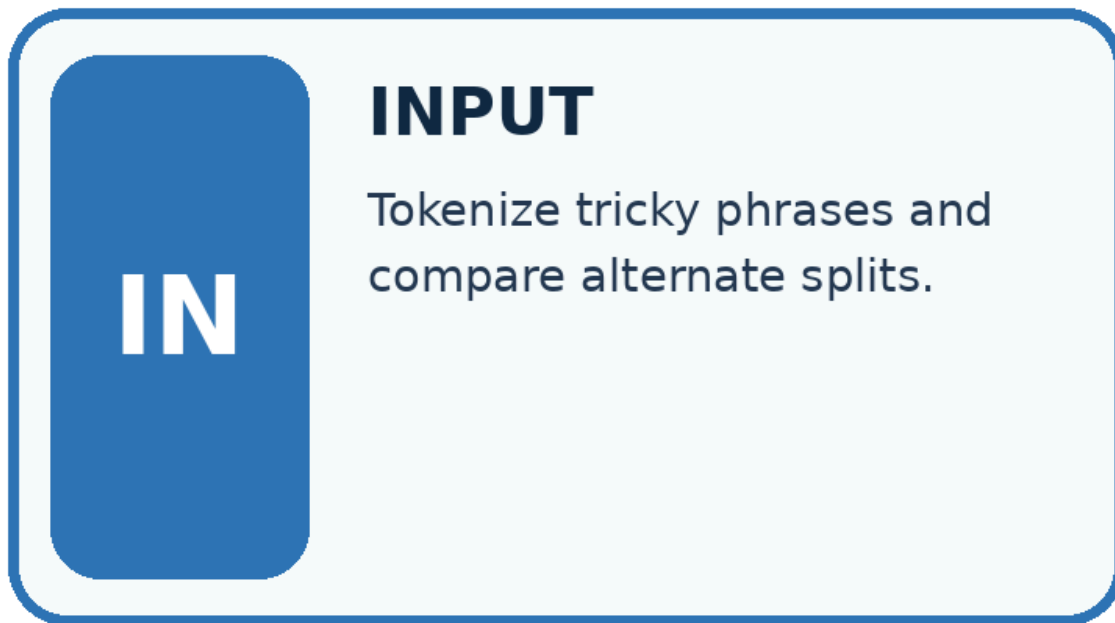
At each position choose the longest matching ordinary token. If no token matches, stop and report the unsupported character and its position. Do not delete the character or guess a replacement. IDs identify entries; a larger ID does not mean a token is more valuable or more similar.

3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓



PROCESS

Language systems process tokens and patterns rather than whole human meanings.



OUT

OUTPUT

Tokenize tricky phrases and compare alternate splits.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Remaining text · Chosen token · ID

Remaining text: cats sat

Chosen token: cat

ID: 1

Remaining text: s sat

Chosen token: s

ID: 5

Remaining text: sat

Chosen token: space

ID: 6

Remaining text: sat

Chosen token: sat

ID: 7

The encoding of cats sat is [1,5,6,7], followed by [0] only if the protocol appends an end token. Decoding concatenates the ordinary token strings to recover cats sat. Count four text tokens, or five sequence items with the end marker included; always state the counting convention. The two written words do not imply two tokens.

Keep the evidence roles separate

Plan or input · Procedure · Permitted conclusion

Plan or input: Exact text: cats sat

Procedure: Longest-match encoding

Permitted conclusion: IDs 1, 5, 6, 7

Plan or input: IDs 1, 5, 6, 7

Procedure: Concatenate token strings

Permitted conclusion: Exact text: cats sat

Plan or input: Exact text: Cat

Procedure: No match for uppercase C

Permitted conclusion: Report unsupported input

Encoding preserves supported characters and makes unsupported input explicit.

Connect the picture and the explanation

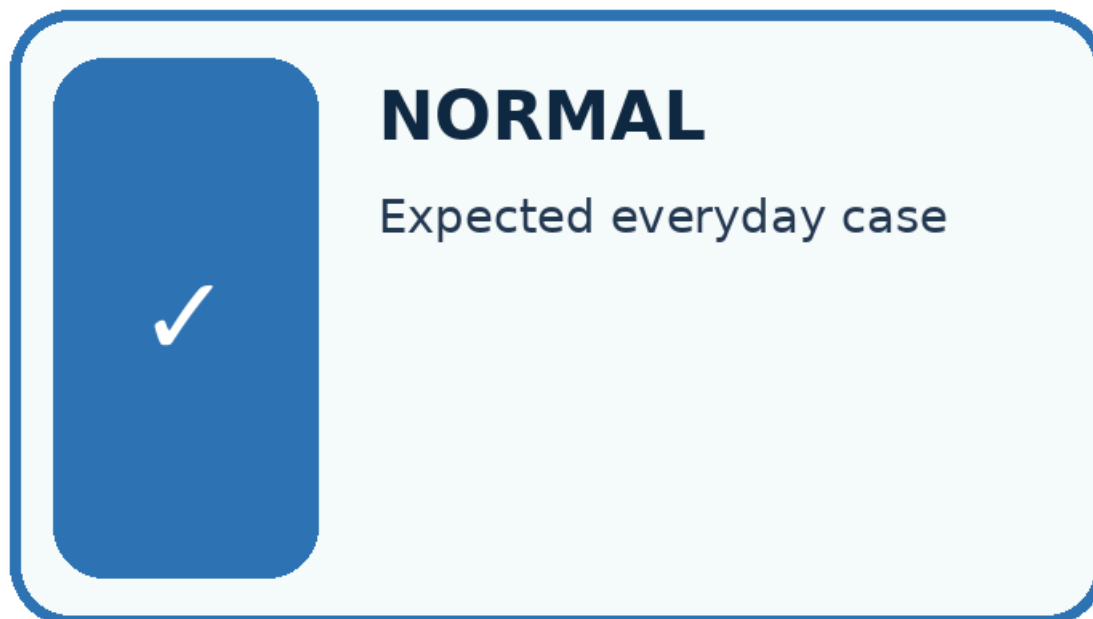
Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

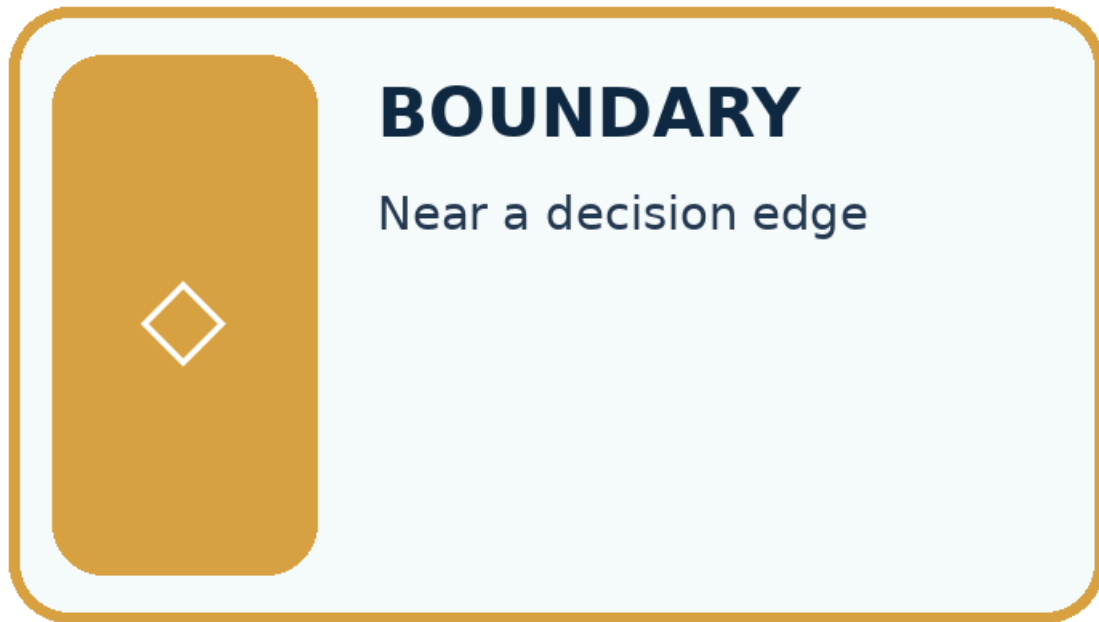
4. Compare the cases

CASE GALLERY

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.

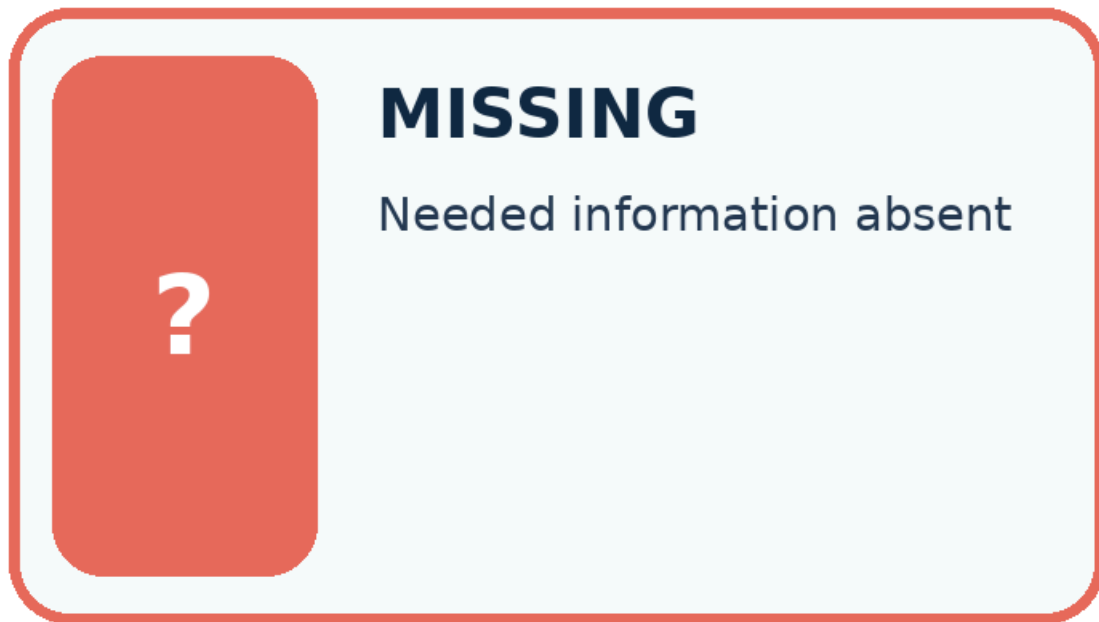




BOUNDARY

Near a decision edge

NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



MISSING

Needed information absent



MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change: Use a character vocabulary covering b,o,k,s and space instead.

Prediction: The same phrase uses ten character tokens.

Reason: There are five letters, one space and four letters; its decoded text remains unchanged.

5. Words that unlock the idea

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

1

TOKEN

piece of text processed by a model

2

TOKENIZATION

rules for splitting text into pieces

3

CONTEXT

surrounding tokens
available

4

PATTERN

repeated relationship
learned from examples

Word · My own example · Not this / boundary

Word: token

My own example: A unit of encoded text processed by a language model, which may be a word, part of a word or punctuation.

Not this / boundary: _____

Word: tokenization

My own example: Converting text into the token units of a particular tokenizer.

Not this / boundary: _____

Word: context

My own example: Information available around a task that helps interpret or answer it.

Not this / boundary: _____

Word: pattern

My own example: A relationship or regularity observed in examples.

Not this / boundary: _____

Words used in the closer explanation

Word · Meaning

Word: Vocabulary

Meaning: The permitted tokens and their identifiers

Word: Encoding

Meaning: Mapping source text into a token sequence under a specified rule

Word: Decoding

Meaning: Mapping a token sequence back to text under a specified rule

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

WEEK 25 • WORKED EVIDENCE

- 1 'unhelpful' might split as 'un' + 'help' + 'ful'.
- 2 A name or emoji may use unexpected token pieces.
- 3 Token boundaries need not match whole words.
- 4 Meaning emerges from patterns across context, not from one token acting like a dictionary.

1. 'unhelpful' might split as 'un' + 'help' + 'ful'.
2. A name or emoji may use unexpected token pieces.
3. Token boundaries need not match whole words.
4. Meaning emerges from patterns across context, not from one token acting like a dictionary.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

A character-only vocabulary `c,a,t,s,space` could encode `cats sat` as `c,a,t,s,space,s,a,t`: eight text tokens. The original vocabulary uses four. Both decode to the same source text. Sequence length is a property of the text, vocabulary and encoding rule together. Counts from different tokenizers are not automatically comparable.

In this vocabulary, `sat` is chosen over `s` because it is longer. The toy longest-match rule is defined explicitly. Real tokenizers may use different algorithms, byte encodings and special-token policies. Never infer an unknown tokenizer's count from word count alone.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Text as Pieces

A fictional tokenizer converts a short catalogue phrase into tokens.

The facts we will use

Vocabulary entries: book, books and the space character. At each position choose the longest matching entry; preserve the actual space.

Text is “books book”. Exclamation marks have no token; zero-based character positions are used.

AI ACADEMY / AI-2/25

Text as Pieces

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Read prefix

At position 0, longest matching entry is books, not book.



2. Encode phrase

Tokens: books, space-character, book; three tokens.



3. Decode

Joining token text returns exactly books book.



4. Unknown character

For books book!, the ! at position 10 has no token: stop with error.

CHANGE ONE THING

Use a character vocabulary covering b,o,k,s and space instead.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Read prefix

What happens and why: At position 0, longest matching entry is books, not book.

Step: 2. Encode phrase

What happens and why: Tokens: books, space-character, book; three tokens.

Step: 3. Decode

What happens and why: Joining token text returns exactly books book.

Step: 4. Unknown character

What happens and why: For books book!, the ! at position 10 has no token: stop with error.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

A token is defined by the vocabulary and encoding rule; it need not be one word or one character. Longest matching matters where vocabulary entries overlap. Counting text tokens here does not count letters. Preserve spaces during decoding so the reconstructed text matches the original exactly.

What this example does not establish

This deterministic toy tokenizer omits real subword training, Unicode handling and model-specific special tokens.

Read the labelled picture: Why is the unknown ! at position 10 rather than position 9?

Change one thing: Use a character vocabulary covering b,o,k,s and space instead. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



COMPARE • DIAGNOSE

Weak idea: Every token is exactly one English word. Strong move: locate evidence, change one cause, and retest.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: works

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: stated rule

Actual: _____

Likely cause / next test: _____

Evidence record: missing input

Expected: safe response

Actual: _____

Likely cause / next test: _____

Evidence record: fresh transfer

Expected: generalizes

Actual: _____

Likely cause / next test: _____

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Decoding round trips check representation, not factual meaning. A supported spelling can be false; an unsupported character can be important. Record the vocabulary version, rule and special-token convention when comparing counts or reproducing model inputs. Whitespace, case and punctuation are part of this exact toy input contract.

Exit: a program drops ! from cat! and announces a successful round trip. Identify the changed information and the honest response.

8. Practise together

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



TRACE THE LOGIC

Predict the result. Trace every condition or calculation. Compare expected with actual. Explain the first difference.

SET evidence_record PREDICT expected_result RUN one normal case and one boundary case
COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST
EXPLAIN result, limitation, and next safe step

Trace one input. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Supported case: cat encodes [1]; c at encodes [2,6,3,4]. A space changes both the character sequence and the possible matches. The input Cat is unsupported at position 0 because uppercase C has no entry. The input cat! encodes cat, then fails on ! at position 3; a partial encoding must not be reported as complete success.

Guided: encode c at, count its text tokens and explain how cat! should be reported.

Practise with less help: Encode and decode

Use workbook W2 for the connected example “Tokenisation and vocabulary”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Tokenisation and vocabulary. Proposed explanation: “A token always equals a word”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Why is the unknown ! at position 10 rather than position 9?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



PREDICT → TEST → RECORD → EXPLAIN

Cut sentence cards into two plausible tokenizations, compare piece counts, remove context, and predict which continuation becomes less certain.

Cut sentence cards into two plausible tokenizations, compare piece counts, remove context, and predict which continuation becomes less certain.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

Test / input: normal

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



MASTERY LADDER
1 Name it • 2 Trace it • 3 Apply it • 4 Challenge it • 5 Explain limits

1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or evidence path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

11. Connect your project

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



TRANSPARENT BOOK RECOMMENDER

Tokenize book descriptions before any text-based matching, while preserving the original source text.

THIS WEEK'S CHECKPOINT Tokenize book descriptions before any text-based matching, while preserving the original source text.

SYSTEM MAP Reader need → allowed inputs → transparent rule or score → ranked candidates → explanation → test record → human decision

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under human control?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

Document the recommender’s text representation and unknown-token behavior rather than assuming every model counts words identically.

12. Show what you know

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



INDEPENDENT PROOF

Tokenize tricky phrases and compare alternate splits.

CHALLENGE Tokenize tricky phrases and compare alternate splits.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My evidence that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

13. Study a complete case

Week 25 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Toy tokenizer A splits rainbows as [rain][bow][s]; B uses [rainbows].

The question

Count tokens and state the limit of the example.

Worked explanation

Three versus one. Token count depends on tokenizer; tokens need not equal words or letters.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

One word can have different token counts

Evidence or stage · Value or result

Evidence or stage: Tokenizer A

Value or result: rain / bow / s

Evidence or stage: Tokenizer B

Value or result: rainbows

Toy tokenizer A splits rainbows as [rain][bow][s]; B uses [rainbows].

Count tokens and state the limit of the example.

Three versus one. Token count depends on tokenizer; tokens need not equal words or letters.

14. Try a new case independently

Independent case: Trace a new input independently

Under the original vocabulary encode sat cats, decode it and count text tokens. Then trace sat cat! and state the exact failure position using zero-based character positions.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 25 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A classroom tokenizer splits sunflower into sun and flower. How many pieces result? Must every real tokenizer split it the same way?

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Tokenize book descriptions before any text-based matching, while preserving the original source text.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

Toy tokenizer A splits 'unhelpful' into un/help/ful; B keeps it whole. Count tokens for each and explain why a three-word sentence need not use three tokens.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: A fictional tokenizer converts a short catalogue phrase into tokens.

By the end, I can · How I will show it

By the end, I can: Encode and decode text under an explicit toy vocabulary

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Explain how segmentation changes sequence length

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Handle unsupported characters without silently deleting information

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Encode and decode text under an explicit toy vocabulary

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.

17. Supported practice, repair and portfolio

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional tokenizer converts a short catalogue phrase into tokens.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? For books book!, the ! at position 10 has no token: stop with error.

- 1. Unknown character
- 2. Decode
- 3. Encode phrase
- 4. Read prefix

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Unknown character For books book!, the ! at position 10 has no token: stop with error. A token is defined by the vocabulary and encoding rule; it need not be one word or one character. Longest matching matters where vocabulary entries overlap. Counting

text tokens here does not count letters. Preserve spaces during decoding so the reconstructed text matches the original exactly.

Practice 2: explain the deciding evidence

Changed-case practice: Use a character vocabulary covering b,o,k,s and space instead. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. We can decide the result without examining the changed input or the procedure.
3. The same phrase uses ten character tokens.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The same phrase uses ten character tokens. There are five letters, one space and four letters; its decoded text remains unchanged.

Repair a missing building block

Review Feedback Loops (ai-2/24). Goal: Implement a deterministic recommender. Short practice: What extra field would reveal why comparing raw clicks is misleading?. Return to this week and explain what you changed.

Review Nearest Neighbor Reasoning (ai-2/22). Goal: Compute absolute distances and rank training neighbours. Short practice: Why do $k=1$ and $k=3$ disagree for the original query?. Return to this week and explain what you changed.

Review Sampling and Representation (ai-2/19). Goal: Define a question and target population. Short practice: Can 80% response be reported as 80% of the whole grade responding?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.